

WealthWise

An Intelligent Portfolio Analytics and Goal-Planning Platform for Indian Mutual Fund Investors

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The Hidden Cost of Financial Blindness



Portfolio Fragmentation

An investor with SBI, HDFC, and Axis funds has no single dashboard showing total portfolio value, overall gain/loss, or combined risk exposure. Each AMC app shows only its own funds — creating dangerous blind spots.



Return Calculations Are Misleading

Platforms show absolute returns (e.g., "+22%") but never XIRR — the only mathematically accurate measure for SIP investors with irregular cashflows. An investor showing +22% absolute may actually have just 8% XIRR over 3 years.



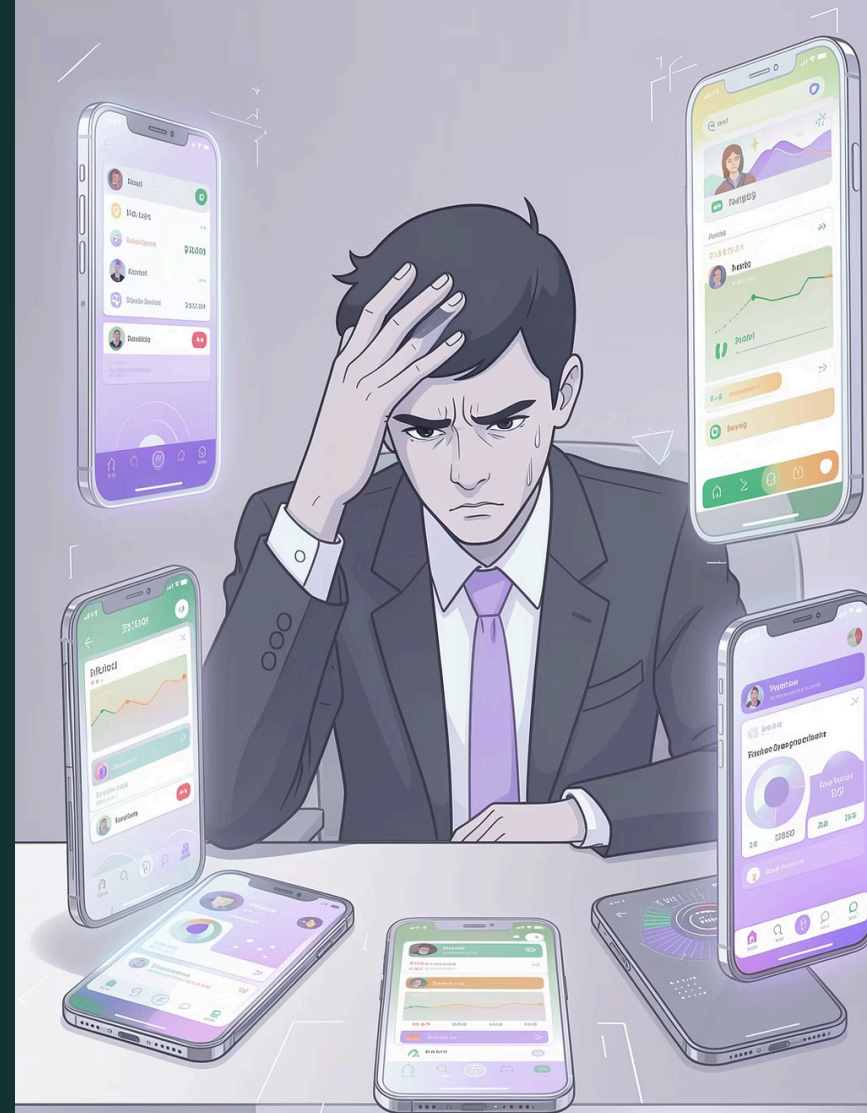
Risk Is Invisible

No free tool tells a retail investor their portfolio's Sharpe Ratio, annualized volatility, or max drawdown. Without these metrics, investors unknowingly hold portfolios that take excessive risk for the returns they generate



Goal Planning Uses Fantasy Math

"Invest ₹10,000/month at 12% for 20 years = ₹1 Crore" — this flat-rate projection ignores market volatility entirely. In reality, outcomes range from ₹60L to ₹1.8Cr depending on sequence of returns.e.



Intelligent Solution: Your Mutual Fund Command Center

WealthWise is a production-grade full-stack platform that consolidates all mutual fund investments into one intelligent dashboard, delivering real-time valuations, institutional-grade analytics, probabilistic goal planning, and automated tax tracking.



01

Fragmented Data

Unified portfolio from CSV import + manual entry across any AMC

03

Invisible Risk

Sharpe Ratio, Volatility, Max Drawdown, SEBI Riskometer (1-6)

05

Tax Leakage

Automated FIFO lot tracking with STCG/LTCG split on every redemption

Technology Stack: React 19 SPA + Spring Boot 3.2 REST API + PostgreSQL 15 (Supabase) + Render Cloud

02

Misleading Returns

Newton-Raphson XIRR engine with 1e-7 precision per fund

04

Fantasy Goal Planning

10,000 Monte Carlo simulations with inflation-adjusted P10/P50/P90

06

Stale NAV Data

4-level NAV fallback: DB → Scheme Master → Live MFAPI → Cost basis

Core Capabilities



Multi-Source Transaction Import

- CSV bulk import: 6 date formats, 11 transaction types
- Manual form with auto-NAV lookup
- Auto-folio generation with scheme resolution
- 45,000+ AMFI scheme database



Real-Time Portfolio Valuation

- Holdings aggregated from investment_lots
- 4-level NAV fallback chain
- Portfolio totals with overall gain/loss
- Monthly growth timeline from NAV history



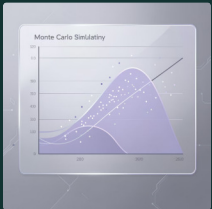
XIRR & Returns Engine

- Newton-Raphson (1,000 iter, 1e-7 tolerance)
- Bisection fallback for edge cases
- Absolute Return + CAGR calculations
- Per-fund: units, cost, current value, gain/loss



Risk Analytics Dashboard

- Sharpe Ratio: $(\text{Return} - 7\% \text{ RFR}) / \text{Volatility}$
- 12-category weighted volatility
- SEBI Riskometer (1-6), Max Drawdown
- Diversification Score (0-10)



Monte Carlo Goal Engine

- 10,000 Gaussian simulations per goal
- P10/P50/P90 in today's rupees
- Required SIP calculator
- 3 sensitivity scenarios

Intelligent Features

1

FIFO Tax Automation

- BUY creates InvestmentLot
- SELL iterates oldest lots first (FIFO)
- ≥ 365 days = LTCG, else STCG
- Auto-computed per transaction

2

SIP Intelligence

- SIP vs Lumpsum comparison per fund
- Rupee-cost averaging quantification
- Advantage ranking across portfolio

3

45K Scheme DB + Live NAV

- Nightly AMFI NAVAll.txt sync
- pg_trgm fuzzy search
- On-demand MFAPI live fetch

4

Notifications & Alerts

- In-app alerts (Info/Watch/Action)
- Email notifications via Gmail SMTP

5

Profile & Settings

- Profile: name, PAN, DOB, nominee
- OTP-based password reset
- Dark theme UI

End-to-End Data Flow



Core Engineering Challenges



Multi-Strategy NAV Resolution

Problem: NAV data from multiple sources with varying freshness. A failed API call shouldn't break the portfolio view.

Solution: 4-level fallback in `HoldingsService.getHoldings()`: `nav_daily` → `scheme_master` → Live MF API (8s timeout, cached back) → avg cost NAV. Stale detection: >5 days triggers refresh; >1 year = discontinued, skip.



XIRR Convergence on Real-World Data

Problem: 50-120+ irregular cashflows. Newton-Raphson can diverge with bad guesses or zero derivatives.

Solution: Triple-strategy in `XirrService`: Newton-Raphson (guess=0.1) → Newton-Raphson (guess=-0.5) → Bisection [-0.999, 5.0]. NaN/Infinity guards, short-period bypass (<7 days), sanity clamp [-99%, +500%].



Concurrent NAV Upsert Race Conditions

Problem: Multiple requests for same scheme may INSERT the same (scheme_code, nav_date) row simultaneously.

Solution: PostgreSQL UNIQUE constraint + app-level `existsBySchemeCodeAndNavDate()` + try-catch silently ignoring duplicate key. First writer wins; others proceed with fetched value.



CSV Import Flexibility

Problem: Users export CSVs from different platforms with inconsistent headers, dates, and types.

Solution: `coalesce()` tries 5+ header variants per field. 6 date parsers. Type normalizer maps 30+ aliases → 11 canonical types. RFC 4180 CSV parser with quoted fields.

Algorithms & Intelligence

1

Newton-Raphson XIRR

Used In: XirrService

Details: $NPV(r) = \sum [C_i / (1+r)^{(t_i/365.25)}] = 0$. Iterative: 1,000 max iterations, $1e-7$ tolerance. Bisection fallback for convergence issues.

2

Monte Carlo Simulation

Used In: GoalEngineService

Details: 10,000 Gaussian simulations. $r = \mu + \sigma \times N(0,1)$ with volatility floor. P10/P50/P90 inflation-deflated projections.

3

FIFO Lot Consumption

Used In: TransactionController

Details: Oldest lots consumed first. Holding $\geq 365d$ = LTCG, else STCG. Per-lot gain computed automatically.

4

Sharpe Ratio

Used In: AnalyticsService

Details: Sharpe = (Return - 7% RFR) / Volatility. Category-weighted across 12 MF categories for comprehensive risk assessment.

5

Required SIP Calculator

Used In: GoalEngineService

Details: Inflation-adjust target $\rightarrow$ subtract FV of corpus $\rightarrow$ solve annuity for SIP. Includes extra months calculator for flexibility.

6

SIP vs Lumpsum

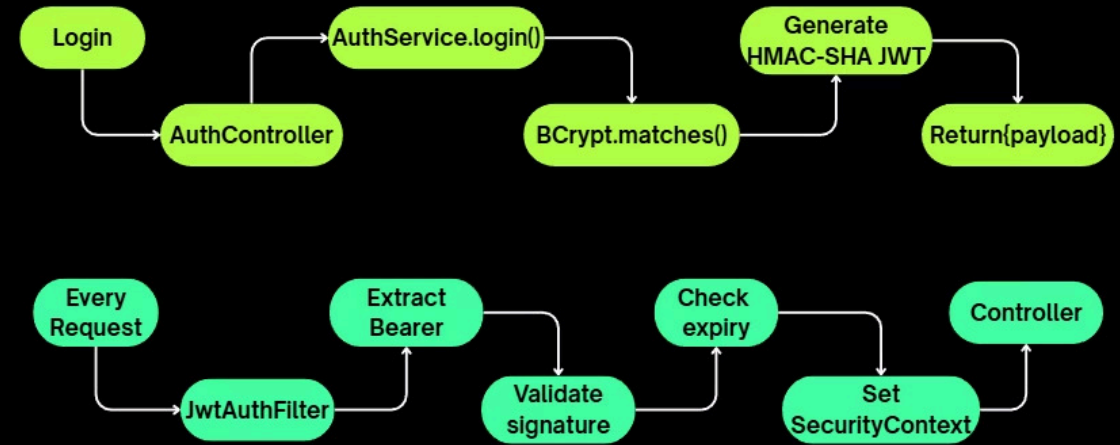
Used In: AnalyticsService

Details: Actual multi-lot DCA value vs hypothetical day-1 lumpsum. Ranked by advantage to optimize investment strategy.

Security & Authentication Pipeline



Authentication Pipeline:



Security Measures

Password Theft Protection

BCrypt adaptive hashing (Spring Security PasswordEncoder) with salt rounds prevents rainbow table attacks

Session Hijacking Prevention

Stateless JWT eliminates server-side session storage, reducing attack surface significantly

Token Forgery Defense

HMAC-SHA256 signed with 54-character secret ensures message integrity and authenticity

Brute Force Reset Protection

SecureRandom OTP via Gmail SMTP with rate limiting prevents automated attacks on password reset

Cross-Origin Attack Prevention

Configurable CORS whitelist restricts API access to authorized domains only

SQL Injection Prevention

JPA parameterized queries only, never string concatenation for database operations

Transport Security

PostgreSQL sslmode=require encrypts all data in transit with TLS 1.2+

Secret Exposure Prevention

Environment variables on Render cloud, never hardcoded in source code or version control

Scalability



Docker Optimization

Leveraging multi-stage builds for a **200MB JRE-only image** (vs. ~800MB JDK), significantly reducing deployment size and startup times.



High-Performance Caching

Caffeine cache with 500 entries and a 10-minute TTL, enhancing responsiveness for frequently accessed data.



Robust Connection Pool

HikariCP configured with 5 max connections, 60s keepalive, and health-check SQL, ensuring efficient and reliable database interactions.



NAV Deduping Strategy

PostgreSQL **UNIQUE constraint** combined with app-level checks prevents race conditions and duplicate entries during NAV upserts.



Stateless JWT Authentication

Enables **horizontal scaling** by eliminating server-side sessions, where JWTs are HMAC-SHA256 signed with a 54-character secret for authenticity.



Enhanced Security

Includes **SecureRandom OTP** for password resets, configurable CORS whitelist, JPA parameterized queries, **PostgreSQL SSL encryption**, and environment variables for secret management.

Performance Metrics & Testing

<50ms

XIRR Computation

Calculates for 120+ cashflows, ensuring rapid financial analysis.

<2s

Monte Carlo Simulation

Executes 10,000 simulations over 240 months per goal, delivering quick risk assessments.

45K+

AMFI Scheme Database

Supports over 45,000 mutual fund schemes, providing comprehensive market coverage.

<500ms

API Response Time

Retrieves user holdings in under 500ms with intelligent caching, enhancing user experience.

~200MB

Optimized Docker Image

JRE-only build reduces image size, enabling faster deployment and lower resource consumption.

8s

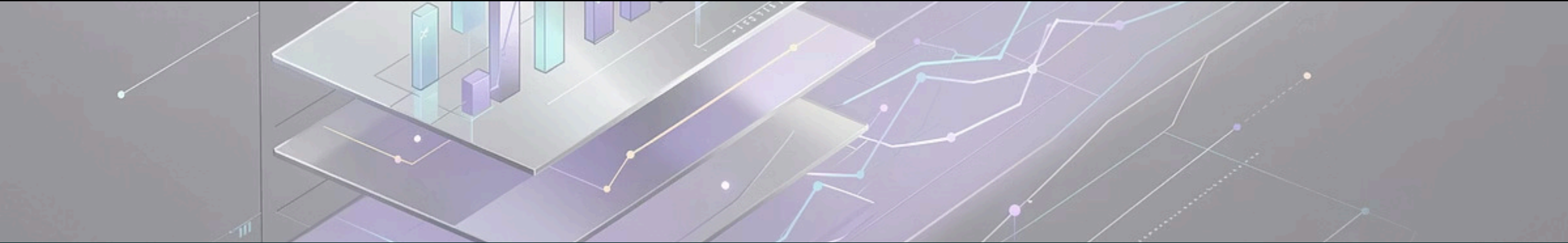
NAV Live Fetch

API calls for Net Asset Value data are designed with an 8-second timeout and robust caching.

Testing Architecture

Our robust testing architecture ensures the reliability and integrity of the WealthWise platform, covering every layer from individual services to full end-to-end user flows.

Layer	Tool	Test Suite	Key Verifications
Service	JUnit 5 + Mockito	AuthServiceTest	BCrypt hashing, JWT generation, password change logic
Service	JUnit 5	XirrServiceTest	Newton-Raphson convergence, edge cases for XIRR calculation
Service	JUnit 5	GoalProjectionServiceTest	Monte Carlo simulations, deterministic projections, required SIP calculations
Controller	MockMvc	AuthControllerTest	HTTP status codes, authentication flow responses
Controller	MockMvc	HoldingsControllerTest	Holdings retrieval, NAV fallback mechanisms
Frontend	Vitest + RTL	App.test.jsx	Router navigation, authentication guards, component rendering
Frontend	Vitest	goalHelpers / mfApi	Helper function logic, API utility correctness
E2E	Playwright	auth / dashboard / goals	Comprehensive user flows for authentication, dashboard interactions, and goal management



Conclusion: WealthWise Achievements

1

Democratized Analytics

Institutional-grade tools like Sharpe Ratio, XIRR, and Monte Carlo simulations are now free and accessible to all retail investors.

2

Production-Ready Engineering

Robust architecture includes a 12-table schema, FIFO lot tracking, 4-level NAV fallback, advanced Monte Carlo, multi-stage Docker, and JWT security.

3

Solving a Real Problem

WealthWise addresses the ₹50 Trillion market gap for 50M+ Indian MF investors, providing unified portfolio intelligence and goal analysis.



Thank You

WealthWise Team 8 · Java Full Stack · VTU Internship 2026

GitHub: github.com/Faishal058/VTU_INTERN_2026_Team8_JAVA